

Interactions between network synchrony and the dynamics of neuronal threshold

Avner Wallach and Shimon Marom

J Neurophysiol 107:2926-2936, 2012. First published 7 March 2012; doi:10.1152/jn.00876.2011

You might find this additional info useful...

This article **cites** 31 articles, 13 of which can be accessed free at:

<http://jn.physiology.org/content/107/11/2926.full.html#ref-list-1>

Updated information and services including high resolution figures, can be found at:

<http://jn.physiology.org/content/107/11/2926.full.html>

Additional material and information about *Journal of Neurophysiology* can be found at:

<http://www.the-aps.org/publications/jn>

This information is current as of June 1, 2012.

Interactions between network synchrony and the dynamics of neuronal threshold

Avner Wallach and Shimon Marom

Network Biology Research Laboratory, Faculty of Electrical Engineering Technion–Israel Institute of Technology, Haifa, Israel

Submitted 27 September 2011; accepted in final form 2 March 2012

Wallach A, Marom S. Interactions between network synchrony and the dynamics of neuronal threshold. *J Neurophysiol* 107: 2926–2936, 2012. First published March 7, 2012; doi:10.1152/jn.00876.2011.—Synchronous activity impacts on a range of functional brain capacities in health and disease. To address the interrelations between cellular level activity and network-wide synchronous events, we implemented in vitro a recently introduced technique, the response clamp, which enables online monitoring of single neuron threshold dynamics while ongoing network synchronous activity continues uninterrupted. We show that the occurrence of a synchronous network event causes a significant biphasic change in the single neuron threshold. These threshold dynamics are correlated across the neurons constituting the network and are entailed by the input to the neurons rather than by their own spiking (i.e., output) activity. The magnitude of network activity during a synchronous event is correlated with the threshold state of individual neurons at the event's onset. Recovery from the impact of a given synchronous event on the neuronal threshold lasts several seconds and seems to be a key determinant of the time to the next spontaneously occurring synchronous event. Moreover, the neuronal threshold is shown to be correlated with the excitability dynamics of the entire network. We conclude that the relations between the two levels (network activity and the single neuron threshold) should be thought of in terms that emphasize their interactive nature.

neuron; response clamp; burst

SYNCHRONY IS A MOST BASIC mode of activity in the cortex, believed to be critical for a range of functional brain capacities (Singer 1999). In most cases, regardless of the complex topology of connectivity (e.g., Eytan and Marom 2006; Bonifazi et al. 2009), the cortical synchronous event is considered a collective population phenomenon, emerging from interactions among the elements comprising the network (e.g., Abeles 1991; Beggs and Plenz 2003; Eckmann et al. 2010), rather than being driven by a uniquely identified class of pacing neurons. To address the generation of network-wide synchronous events, as well as the factors constraining the frequency of their occurrence, simultaneous monitoring of relevant state variables at both the network and cellular levels is wanted.

While an ultimate design aiming at such multilevel monitoring would involve measurement and stimulation in the intact brain, obvious issues of experimental stability and control make the choice of large-scale networks of cortical neurons developing in vitro an attractive alternative in this context (Marom and Shahaf 2002). These networks are readily accessible to well-controlled long-term experiments and exhibit the

dynamics of synchrony that is comparable in some key features to synchrony observed in vivo (Eytan and Marom 2006; Ham et al. 2008; Jimbo et al. 1993; Maeda et al. 1995; Stegenga et al. 2008; Van Pelt et al. 2004; Wagenaar et al. 2006).

Previous experiments that attempted simultaneous monitoring of both single neuron processes and population level activity in vitro clearly indicate interdependency between the two; these studies show correlation of intracellular parameters, such as calcium concentration (e.g., Robinson et al. 1993), membrane potential (e.g., Giugliano et al. 2004), and the occurrence of network synchronous events. Indications to similar relations might also be found in vivo (Li et al. 2009). Following the lead of these and related studies, we sought for a functional link between intracellular factors and network activity. We lump all the relevant processes at the cellular level into one functional indicator: the neuronal threshold. The idea is to monitor threshold dynamics of individual neurons simultaneously with the dynamics of network synchrony. To that end, we implemented a recently introduced technique, the response clamp, that enables online monitoring of single neuron threshold dynamics. The technique is fast and stable enough to monitor threshold dynamics at the single neuron level over extended timescales, while network synchronous activity continues in the background.

In a recent method study (Wallach et al. 2011), we reported in passing that the occurrence of a network synchronous event is accompanied by a biphasic change in the neuronal threshold (see Fig. 7 in that article). The potential significance of this finding as a window into the relations between single neuron and population dynamics led us to focus in the current report on this basic phenomenon, seeking to understand what factors affect it and what are its functional consequences. We show that 1) these threshold dynamics are correlated across the neurons constituting the network; 2) these threshold dynamics are entailed by the input to the neurons rather than by their own spiking (i.e., output) activity; 3) the magnitude of network activity during a synchronous event is correlated with the threshold state of individual neurons at the event's onset; and, finally, 4) recovery from the impact of a given synchronous event on neuronal threshold lasts several seconds and seems to be a key determinant of the timing of the next spontaneously occurring synchronous event, as well as of the network's responsiveness to external stimulation.

MATERIALS AND METHODS

Cell Preparation

Cortical neurons were obtained from newborn rats (Sprague-Dawley) within 24 h after birth using mechanical and enzymatic procedures

Address for reprint requests and other correspondence: A. Wallach, Network Biology Research Laboratory, Faculty of Electrical Engineering Technion-Israel Institute of Technology, Haifa 32000, Israel (e-mail: avnerw@technion.ac.il).

described in earlier studies (Marom and Shahaf 2002). Rats were killed by CO₂ inhalation according to protocols approved by the National Institutes of Health. The protocol was approved by the Inspection Committee on the Constitution of the Animal Experimentation at the Technion, approval number IL-099-08-10. The neurons were plated directly onto substrate-integrated multielectrode arrays and allowed to develop functionally and structurally mature networks over a time period of 2–3 wk. The number of neurons in a typical network is of the order of 1,300,000, covering an area of ~380 mm². The preparations were bathed in MEM (Sigma-Aldrich) supplemented with heat-inactivated horse serum (5%), glutamine (0.5 mM), glucose (20 mM), and gentamycin (10 µg/ml) and maintained at 37°C, 5% CO₂-95% air in an incubator as well as during the recording phases. An array of 60 Ti/Au/TiN extracellular electrodes, 30 µm in diameter and spaced 500 µm from each other [MultiChannelSystems (MCS), Reutlingen, Germany], was used. The insulation layer (silicon nitride) was pretreated with polyethyleneimine (Sigma; 0.01% in 0.1 M borate buffer solution).

Measurements and Stimulation

A commercial amplifier (MEA-1060-inv-BC; MCS, Reutlingen, Germany) with frequency limits of 150–3,000 Hz and a gain of ×1,024 was used. Monophasic square pulse 200-µs, 100- to 1,000-mV voltage stimulation through extracellular electrodes was performed (Wagenaar et al. 2004) using a dedicated stimulus generator (MCS). The distance between stimulating and recording electrodes is usually ~2 mm. Shortly after stimulation direct, antidromically evoked action potentials may be detected by nonstimulating electrodes. The 30- to 60-suprathreshold (500–800 mV) stimuli were given from each electrode. Voltage traces of the first 15 ms following each stimulus were recorded from all electrodes. Direct antidromic responses were distinguished from early, synaptically mediated ones based on their high reliability and precise, low-jitter latency, as described in previous reports (Gal et al. 2010, see also supplementary material in Kermay et al. 2010). The latency of each neuron at the onset of stimulation was defined as the baseline spike latency of this neuron, denoted L_0 (mean ± SD was 5.6 ± 1.8 ms in $n = 16$ neurons presented in this study). This baseline latency, as well as the spike shapes, were used to isolate single unit activity. In this study, we use the term “single neuron activity” to denote spiking activity from such isolated units. Data were digitized using data acquisition board (PD2-MF-64–3M/12H; UEI, Walpole, MA). Each channel was sampled at a frequency of 16 k samples/s. Data processing and closed-loop stimulation were performed using a Simulink (The Mathworks, Natick, MA)-based xPC target application (see Zrenner et al. 2010 for details).

Response Clamp

The response-clamp technique, which enables continuous monitoring of threshold dynamics, is presented in detail in Wallach et al. (2011). Briefly, a neuron that is antidromically responsive to extracellular stimulation is periodically stimulated (typically at 10 Hz) using short (200 µs) pulses. The response probability of the neuron to this stimulation is estimated based on recent response history, and a control algorithm adjusts the stimulation pulse amplitude so that a constant 50% response probability is maintained (see algorithm’s details below). Thus the sequence of pulse amplitudes emitted by the controller is in fact a recording of the changes in the neuronal stimulation threshold, defined as the pulse amplitude at which an action potential is generated with a probability of 50%. The response clamp follows the rationale of the voltage-clamp technique (Hodgkin et al. 1952), where the feedback current injected to control the voltage reflects the intrinsic membrane current and thus analysis of this control signal exposes the internal dynamics of conductance.

Estimating spike probability online. Let s_i be an indicator function, so that $s_i = 1$ if the neuron generated an evoked spike after the i^{th}

stimulus and $s_i = 0$ otherwise. We denote $\tilde{P}_n$ the response probability estimation calculated at time $t > t_n$, based on the set of responses $\{s_1, s_2, \dots, s_n\}$ to stimuli given at times $\{t_1, t_2, \dots, t_n\}$. A weighted average was realized by using the recursive formula

$$\tilde{P}_n = \left(1 - e^{-\frac{t_n - t_{n-1}}{\tau}}\right) \cdot s_n + e^{-\frac{t_n - t_{n-1}}{\tau}} \cdot \tilde{P}_{n-1}. \quad (1)$$

Unless stated otherwise, τ (the estimation time-constant) was set to 10 s. With stimulation frequency of 10 Hz, Eq. 1 gives 1% weight to the last response and 99% to the previous estimation. It can be shown that, assuming the response probability $P(t)$ is constant, the estimator is asymptotically unbiased, namely:

$$\lim_{n \rightarrow \infty} \mathbf{E}[\tilde{P}_n] = \mathbf{E}[s_i] = P, \quad (2)$$

where $\mathbf{E}[x]$ denotes the expected value of x . For a time-varying probability (i.e., when the above mentioned assumption of constant response probability does not hold), the expected estimation is a weighted average (i.e., a low-pass filtered version) of $P(t)$. Thus slower kernels yield smoother estimations while being less sensitive to abrupt changes in responsiveness (see Fig. 8A).

To reduce the chances of identifying a “spontaneous” spike (i.e., one that was elicited by the network and not by stimulation) as a directly evoked one, the time window of response detection in the clamped neuron was set to be $[L_0 - 1 \text{ ms}; L_0 + 3 \text{ ms}]$, where L_0 is the baseline spike response latency (i.e., the latency of response to low rate stimulation). This baseline latency was shown to be a highly reliable, intrinsic measure which is unaffected by synaptic blockade (see Fig. 1 in Gal et al. 2010). The time window encompasses the dynamic range of evoked response latency in these neurons. This procedure also minimizes the risk of detecting spikes from multiple neurons. To verify this, spikes shapes were observed throughout the experiment (for further discussion on single cell isolation in these cultures, see Gal et al. 2010).

Proportional-integral-derivative controller. A proportional-integral-derivative controller was realized on the xPC target system (Levine 1996). The input to the controller is the error signal

$$e_n = P_n^* - \tilde{P}_n \quad (3)$$

where P_n^* and $\tilde{P}_n$ are the desired and estimated response probabilities at the n^{th} stimulus, respectively. The output of the controller is generally composed of four components

$$A_n = A_{\text{baseline}} + g_P e_n + g_I \sum_{i=1}^n e_i + g_D (e_n - e_{n-1}) \quad (4)$$

where g_P , g_I , and g_D are the proportional, integral and derivative gain parameters, respectively (unless stated otherwise, g_P was 400 mV, g_I was 160 mV, and g_D was 0), and A_{baseline} is the baseline amplitude bias (set to 500 mV). In our system maximal stimulation amplitude was 1,000 mV.

Data Analysis

The data were analyzed using Matlab (The Mathworks) and Mathematica (Wolfram Research, Champaign, IL). Action potentials were detected online by threshold crossing. All spike times and shapes, as well as voltage traces from all electrodes 15 ms after each stimulus, were recorded for analyses. Population activity rate was calculated offline by binning the spikes of all the electrodes not involved in the clamp into 10-ms bins, followed by smoothing with a 100-ms sliding window. Burst onset was defined as the time bin in which the population activity rate crossed a threshold (200 Hz) upward, with a refractory period between bursts of 1s. All burst-triggered averages were computed using 100-ms time bins when stimulation rate was 10 Hz (all figures except Fig. 8) and 200-ms time bins when stimulation rate was 5 Hz (see Fig. 8). Only bursts in which the clamp was

effective (i.e., clamp error did not exceed 5% from 10 s before to 20 s after burst onset) were averaged. Due to the large variability of the threshold changes (SD ~ 16 mV, see RESULTS), ≥ 100 bursts were needed in order for the SE to be significantly smaller than the measured signal (around several millivolts in magnitude). When bursts of equal neuronal activity were used, spike sorting was performed to verify that only action potentials of the clamped neuron are counted (see Fig. 4B). This was done using principle component analysis of spike shapes, followed by K-means clustering using the first three components (Lewicki 1998).

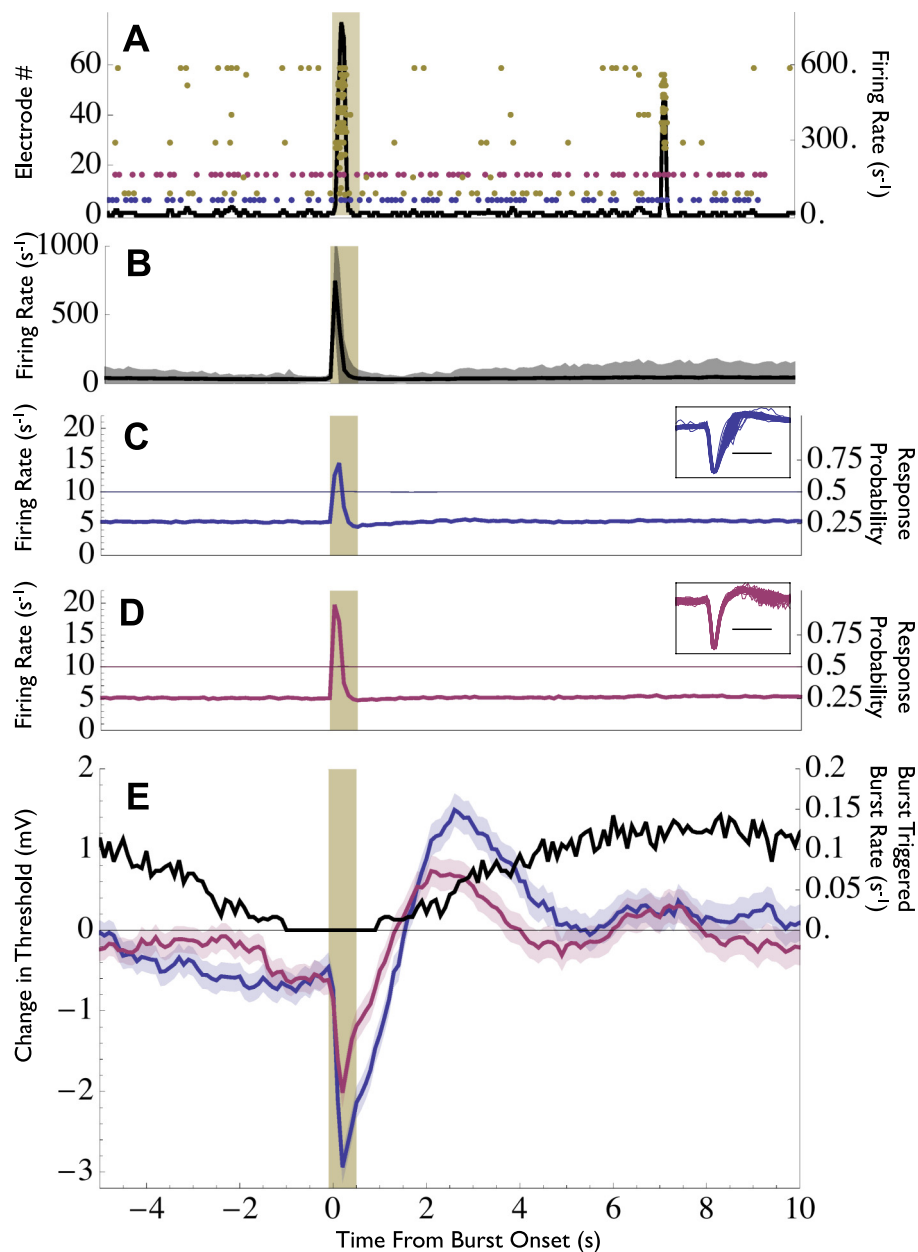
Statistics Across Preparations

The analyses presented in this work were performed on a total of 16 neurons, collected from 11 different cultures. However, some statistical analyses were not applicable over the entire database. Therefore, in each analysis the specific number of neurons and cultures used is quoted. Statistical results across neurons are shown in the form of means $\pm$ SD.

RESULTS

The experiments presented in this work were conducted in cortical neural networks developing in vitro. The activity of these networks is characterized by brief (ca. 0.1–0.3 s) episodes of synchronous events called network spikes or bursts, which occur once every several seconds (Droge et al. 1986; Jimbo et al. 1993; Maeda et al. 1995; Van Pelt et al. 2004; Eytan and Marom 2006; Wagenaar et al. 2006; Stegenga et al. 2008; Ham et al. 2008) (Fig. 1A). This activity was monitored through an array of 60 extracellular electrodes. To continuously measure the threshold of a given single neuron embedded in the network on the background of the population activity, we applied a closed-loop stimulation technique called the response clamp. This method “reports” single neuron threshold dynamics by continuously monitoring the stimulation amplitude required to generate a response with probability of 50% (see MATERIALS AND METHODS). The setup, which enables simultane-

Fig. 1. Threshold fluctuations and network synchrony. **A:** raster plot depicting neural activity while response clamp is performed. Yellow dots mark action potentials detected in all the electrodes that did not respond to the response-clamp stimulation (left-hand ordinate denotes electrode number). Blue and purple dots mark action potentials in the 2 clamped neurons. Thick black line depicts network activity rate (right-hand ordinate). Time axis is relative to a detection of a spontaneous burst (marked with shaded area). Note the additional, smaller burst detected around time 7 s. **B:** burst-triggered average of total network firing rate (total spike rate in all electrodes, save for the ones used for response clamp) in bins of 100 ms (thick black line). Shaded gray area depicts SD. Vertical shaded area signifies the “burst epoch,” from 0.1 s before threshold crossing to 0.5 s after threshold crossing. **C** and **D:** burst-triggered averaged activity of the two clamped neurons (bin size = 0.1 s). Thick lines show the firing rate (left-hand ordinate), while the thin lines depict the response probability (right-hand ordinate). **Insets:** voltage traces of 100 randomly chosen spikes, normalized to their minimum value (calibration bar = 1 ms). **E:** burst-triggered averages of changes in neuronal thresholds, as reflected by the response clamp’s control signal (colored lines, left-hand ordinate), relative to their preburst value (the mean threshold between 10 and 5 s before burst onset). Shaded areas represent the SE. Threshold in both neurons exhibits a biphasic transient, first sharply dropping, then surpassing its preburst value, and finally relaxing back towards baseline. Relaxation time is similar in both neurons and resembles the kinematics of the burst-triggered burst rate of the network (thick black line, right-hand ordinate). This rate was calculated by dividing the average number of bursts detected in each time bin (relative to burst onset) by the bin size. Note that the burst rate is zero for ~ 1 s and then recovers slowly (~ 6 s) towards the mean burst rate of the network (0.113 bursts/s).



ous recording of both the dynamics of neuronal threshold and that of network activity, was used to collect large data sets in which thousands of bursts were analyzed to expose their relations to single neuron threshold changes. In the first part of the RESULTS, we present the findings arising from these analyses. In the second part, we discuss various control experiments and analyses that further validate our findings.

Experimental Findings

An example of the basic analysis is presented in Fig. 1. Two neurons in the same culture were concurrently clamped for 15 h, during which 6,118 bursts occurred in the network (Fig. 1A). The total network activity in a time window around each of these bursts was used to produce the burst-triggered averaged population firing rate (Fig. 1B, black line). Figure 1, C and D (blue and purple thick lines, related to the left hand ordinate axes), depicts the burst-triggered averaged firing rate of the two clamped neurons (*insets*: sample voltage traces from each neuron). Clearly, the firing rate in each of these two neurons is dramatically modulated during the network synchronous event. However, the response probability of these two neurons, clamped by the closed loop algorithm to a value of 0.5, remains stable (thin lines, *right-hand ordinates*), allowing us to extract the dynamics of their thresholds surrounding and during the synchronous burst. The average change in the thresholds of these two neurons in this time window, relative to their preburst value, is presented in Fig. 1E. Note that the response-clamp measurements are relatively noisy (average SD of 15.79 mV in the two neurons presented), reflecting both the limits of the algorithm as well as the steepness of the input-output function of the neuron (Wallach et al. 2011). Thus the averaging over many bursts was essential to improve the signal-to-noise ratio. The SE is presented as confidence bounds in Fig. 1E.

Figure 1E presents our key finding; that is, that network synchronous activity generates a biphasic, long-lasting deflection of neuronal threshold: immediately at burst onset, the threshold decreases, i.e., the neuron becomes hypersensitive (negative peak was -4.97 ± 4.21 mV in $n = 13$ neurons, collected from 8 cultures). This phase is followed by an increase in the threshold, surpassing its preburst value (positive peak was 3.13 ± 3.01 mV in the same 13 neurons), after which it slowly relaxes back towards baseline. While the first, fast phase raises many interesting questions (e.g., its precise onset relative to the neuron's recruitment time within the burst, see Eytan and Marom 2006), the fine details of this phase are beyond the time-resolution of the response clamp (i.e., 100 ms at 10-Hz stimulation rate). We therefore focus our attention in this study on the relaxation process of the second phase. The timescale of this relaxation seems to be similar across different neurons and also resembles the recovery time of the network activity, as reflected in the burst-triggered averaged burst rate (which is related to the burst process autocorrelation function, black line in Fig. 1E). These findings suggest that some relations exist between the threshold status of individual neurons and the population dynamics. Two observations that are demonstrated in Fig. 1 suggest that these threshold dynamics are shared by many (if not most) of the neurons in the network: 1) the timescale of threshold relaxation of an arbitrarily chosen neuron matches the timescale of relaxation of the network-

level burst-rate process, and 2) that average threshold dynamics of two arbitrarily chosen neurons are correlated.

The above-mentioned correlations between threshold dynamics and burst-rate dynamics are best demonstrated by comparing the measurements obtained from different preparations. Figure 2 shows the results obtained from 13 different neurons in a total of 8 cultures. For each culture, we defined the burst rate recovery time as the time at which the burst-triggered burst rate (e.g., the black line in Fig. 1E) reaches 50% of the mean burst rate of that network (i.e., its asymptotic value). The results in Fig. 2 suggest that both the magnitude of deflection (Fig. 2A) and the time course of recovery (Fig. 2B) scale with the characteristic timescale of the network. Contrary to what one might expect, this scale invariance does not seem to reflect similar scaling of spiking activity during the bursts, either of the clamped neuron or of the network (Fig. 2C).

The impact of network synchronous events on the state of single neurons is also reflected in the properties of the spikes generated by these neurons. Figure 3 depicts the burst-triggered averages of changes in spike latency (the time delay between stimulus onset and the detection of an evoked spike; Fig. 3A) and changes in the detected spike amplitude (Fig. 3B) of a neuron, relative to their preburst values. These changes could, of course, be measured without applying the response-clamp technique by simple repetitive suprathreshold stimulation. The changes in neuronal threshold are also shown for comparison (Fig. 3C). As in the case of neuronal threshold, the dynamics of the spike properties indicate an initial increase in excitability (reflected in a shorter latency and bigger spikes), followed by an episode of depression (longer latencies and smaller spikes) that relaxes back to baseline. The timescale of these effects on the spike properties, however, is shorter than that of the threshold dynamics. While the burst impact on these spike properties subsides within a second after burst onset (and well before the next burst occurs), the more subtle effects on the threshold persist throughout the interburst interval.

We now turn to explore the relationships between the threshold dynamics and different aspects of the population activity, starting with the impacts of burst magnitude on threshold deflection: an underlying assumption in this analysis is that the total number of spikes emitted by the network, recorded through the electrode array, is indicative of the magnitude of the synaptic input driving the neuron during the burst. We operationally define the network burst magnitude as the total number of spikes detected in all the electrodes (not including the ones used for the clamp procedure) during the burst (from 0.1 s before, to 0.5 s after burst onset). All the bursts detected during a 20-h response-clamp experiment (total of 14,313 bursts) were divided into two subsets: strong bursts (where burst magnitude was above the median) and weak bursts (where burst magnitude was below the median). The neuronal threshold dynamics in these two burst subsets are shown in Fig. 4A: threshold deflections are more pronounced at the strong bursts (red). All neurons studied exhibited such dependency (difference in peak positive deflection between strong and weak bursts, marked with arrowheads, of 2.79 ± 1.14 mV in $n = 8$ neurons, collected from 5 different cultures). This dependency is apparent throughout the range of burst magnitudes (see Fig. 4A, *inset*).

We note, however, that the network burst magnitude and the single neuron firing rate during the burst are highly correlated;

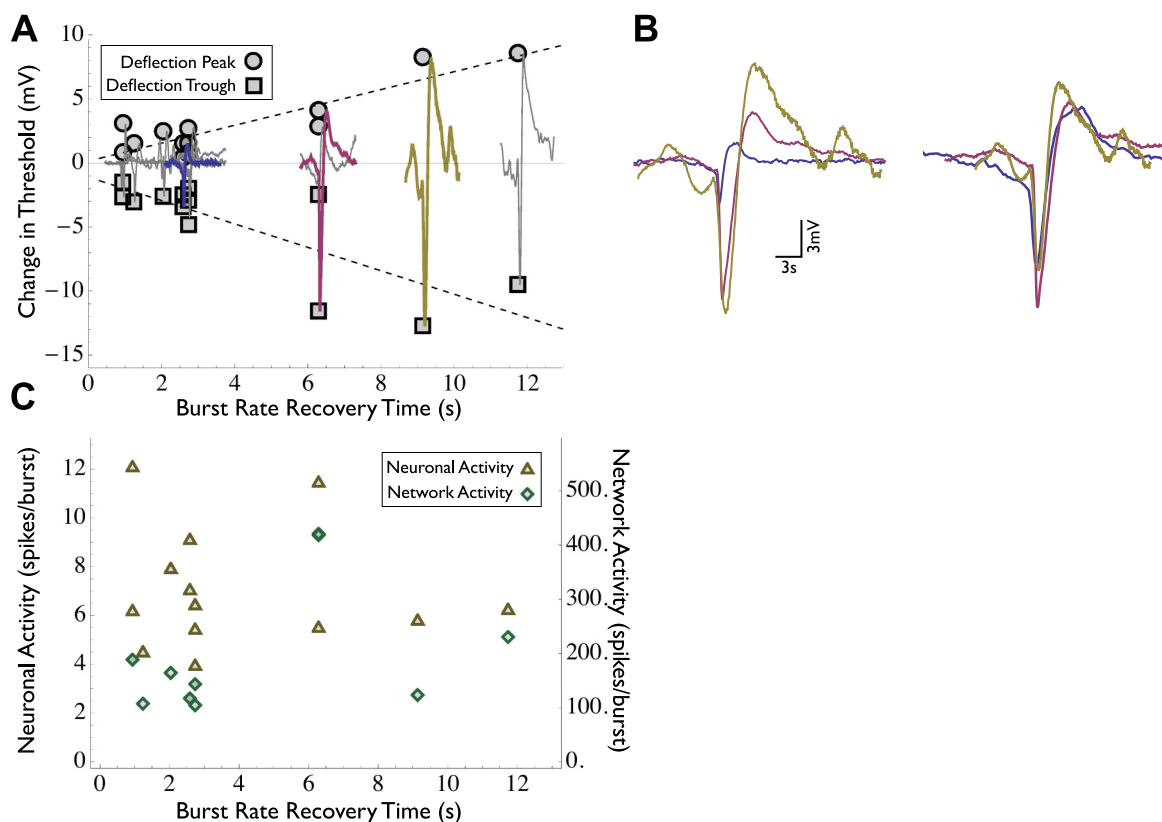


Fig. 2. Correlations between burst rate dynamics and neuronal threshold dynamics. *A*: threshold changes of different neurons ($n = 13$ neurons in 8 cultures), as recorded using identical response-clamp parameters, are presented centered at the burst rate recovery time of the network in which they are embedded (defined as the time where the burst-triggered burst rate reaches 50% of the mean burst rate of the network). Deflection magnitude is correlated with burst rate dynamics; dashed lines depict linear regression of deflection peaks (circles, $R^2 = 0.79$) and troughs (squares, $R^2 = 0.64$). *B*: 3 colored examples of threshold dynamics in *A* are superimposed before (left) and after (right) normalization of both the ordinate and the abscissa by the burst rate recovery time. *C*: both the mean neuronal activity (triangles, left-hand ordinate) and the mean network activity (diamonds, right-hand ordinate) during the bursts show no apparent correlations with burst rate dynamics.

in the above-mentioned experiment, the correlation coefficient between burst magnitude, as defined above, and the total number of spikes emitted by the clamped neuron during the burst was 0.62 (correlation coefficient was 0.68 ± 0.17 in $n = 8$ neurons, collected from 5 different networks). Thus it is fair to ask: is it the intensity of the synaptic input per se that impacts on single neuron threshold dynamics or is it the entailed firing intensity (i.e., the neuronal output) of the clamped neuron that brings about the effect shown in Fig. 4*A*? To explore this issue we repeated the analysis of Fig. 4*A* on a subset of bursts with identical output levels, e.g., all the bursts within which the neuron fired precisely the same number of spikes. Figure 4*B* shows one example of such an analysis, where a subset of bursts in which neuronal activity is narrowly distributed was selected (8 spikes per burst, total of 549 bursts), demonstrating that the threshold deflection remains strongly dependent on burst amplitude even when the output is virtually identical. This analysis was repeated throughout the range of output activity levels, yielding similar results (average difference in peak deflections in this neuron was 1.47 mV across all output levels. In $n = 7$ neurons, collected from four different networks, the difference was 1.63 ± 0.68 mV). The long-term effects of the burst on the single neuron threshold are therefore mostly affected by the neuron's input, rather than by its own spiking activity.

Close inspection of the trajectories presented in Fig. 4*B* reveals an additional point of potential interest; it concerns the threshold level at the burst onset, that is just before the burst occurred (marked with asterisks). Note that strong network bursts were preceded by a decline in the threshold of the clamped neuron (i.e., hypersensitization of the neuron), while weak bursts were preceded by an elevation of this threshold (i.e., desensitization of the neuron). Similar results were obtained in other levels of neuronal output; the average difference between the onset threshold of strong and weak bursts in the presented neuron, across all output levels, was 2.28 mV. This result was consistent in all neurons analyzed, with difference ranging from 0.14 to 4.51 mV (means $\pm$ SD were 2.64 ± 1.75 mV in $n = 7$ neurons, collected from 4 different networks). This implies that the threshold status in a randomly selected neuron, before the emergence of network synchrony, is highly correlated with the activity intensity of the entire network during the burst.

It is thus tempting to hypothesize that the generation of network synchrony (and not only its magnitude) may be related to threshold dynamics, which are correlated amongst different neurons. To check this hypothesis, we performed the analysis shown in Fig. 5*A*, where threshold dynamics for subsets of bursts, which were followed by different durations of quiescence (i.e., no burst occurrence), are presented. This analysis enabled us to observe the threshold dynamics leading to burst

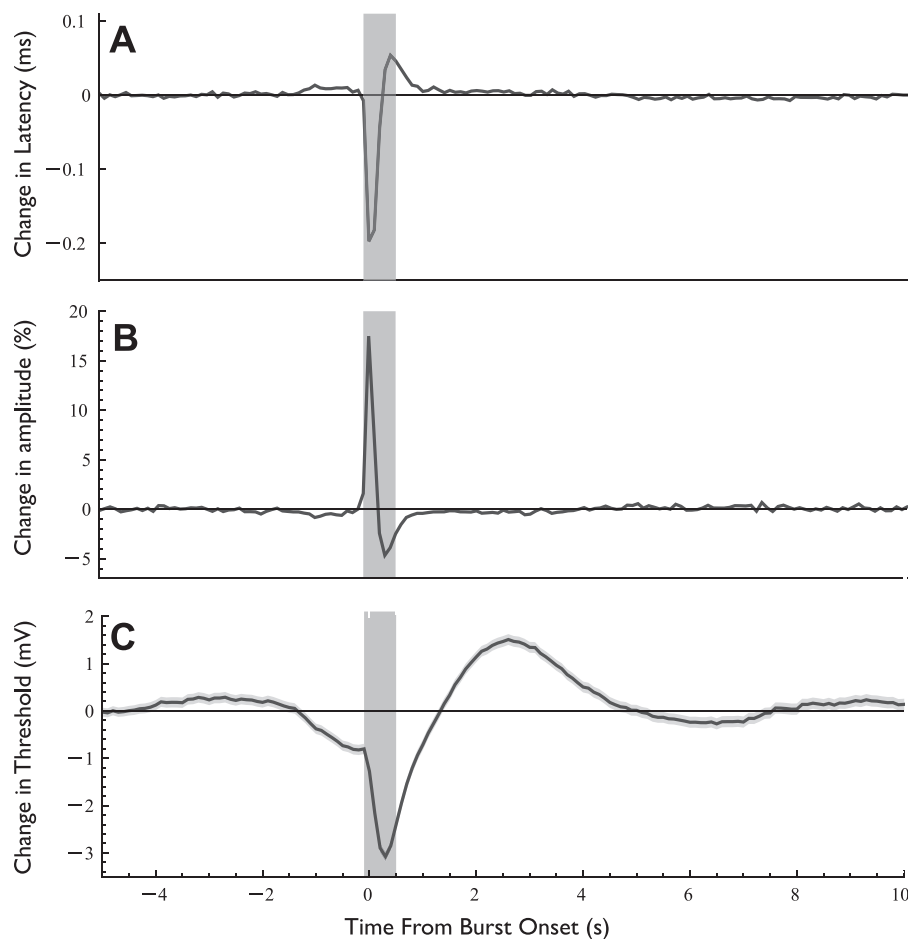


Fig. 3. Effects of network synchrony on spike characteristics. A neuron was clamped for 20 h during which 14,313 bursts were detected. Burst-triggered averages of the change in spike latency (A), spike amplitude (B), and neuronal threshold (C) were computed relative to their preburst value. Onset of network synchrony (shaded area) is accompanied with a sharp decrease in spike latency and an increase in spike amplitude. Later, the effects are reversed and the neuron goes through a short epoch of long latencies and small amplitudes before returning to baseline. Note the difference in timescales between the burst effects on the spike properties and their effects on the neuronal threshold. SE is presented in A–C, yet it is of negligible proportions due to the large data set.

initiation. Figure 5A shows strong correlations between the magnitude and duration of the threshold deflection due to a given burst and the time delay to the following burst. In Fig. 5B, we repeated the analysis of Fig. 5A on many such subsets of bursts (each subset characterized by similar succeeding

interburst intervals); the result of this analysis shows that the time-to-next network synchrony is monotonically increasing with the time of single neuron threshold relaxation from the impact of the previous synchrony. In this network, these monotonic relations saturate ~ 7 s, much like the dynamics of

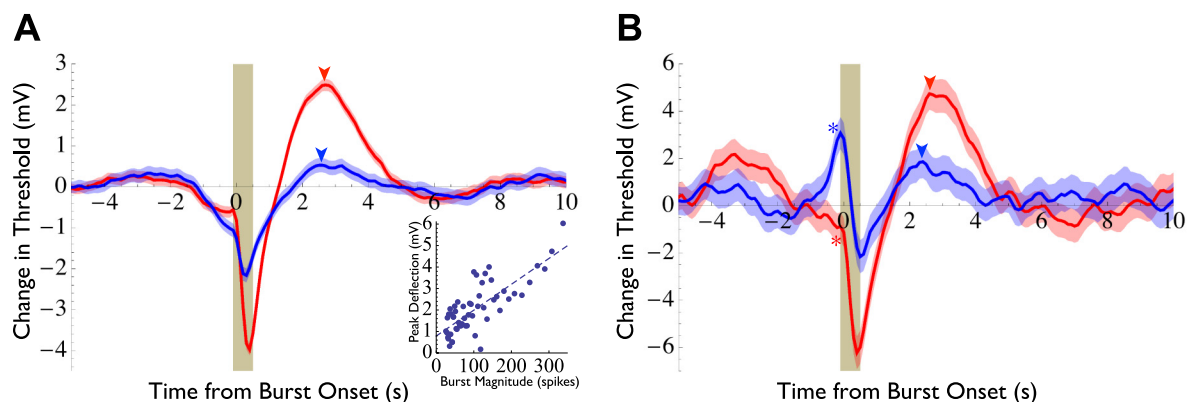


Fig. 4. Effects of network activity intensity on threshold dynamics. A neuron was clamped for 20 h, during which 14,313 bursts were detected. For each burst, the total network activity (total number of spikes in all nonclamped channels, from 0.1 s before until 0.5 s after threshold crossing) and the total neuronal activity in the clamped neuron were computed. Spike sorting was performed on the clamped channel to verify that only the action potentials of the clamped neuron were counted (95% of detected spikes in the channel). A: effects of burst magnitude on threshold dynamics. Entire dataset was divided into 2 subsets, strong bursts (network activity above the median, red) and weak bursts (network activity below the median, blue). Changes in neuronal threshold are markedly amplified in the strong bursts' subset. Inset: bursts were sorted according to their magnitude, in ascending order, and then divided into subsets of 250 bursts each. Peak positive deflection of the averaged threshold in each subset is displayed relative to the mean network activity in that subset (circles). Dashed line displays a least-square fit of the data to a linear equation ($R^2 = 0.61$). B: subset of bursts in which the output of the clamped neuron was identical was selected (8 spikes per burst, total of 549 bursts). This subset was divided into strong input bursts and weak input bursts as in A. Peak of threshold deflection (arrowheads) is higher when the input from the network is strong. Also note the difference in threshold levels at burst onset (*).

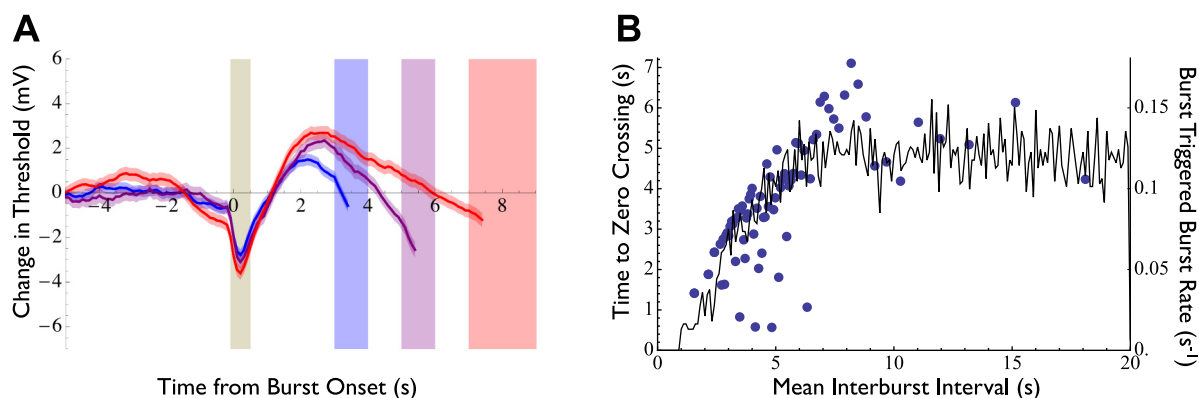


Fig. 5. Threshold dynamics are correlated with onset of subsequent burst. *A*: 3 subsets of bursts were selected: bursts that were followed by an interburst interval (IBI; quiescent period) of 3–4 s (blue), bursts that were followed by an IBI of 5–6 s (purple) and bursts that were followed by an IBI >7 s (red). For each subset, the burst-triggered averaged change in threshold is presented in solid lines (shaded area represent SE) from 5 s before burst onset until the respective region of subsequent bursts (colored vertical shaded areas). Both the magnitude and relaxation time of the threshold deflection are correlated with the time required for the network to generate the next spontaneous burst. *B*: all the bursts were sorted according to the subsequent IBI, in ascending order, and then divided into subsets of 150 bursts each. With the use of the average threshold trajectory, the time of first downward zero crossing was computed for each subset, and displayed relative to the average subsequent IBI in that subset (circles, *left-hand ordinate*). Dependency of the 2 measures is monotonically increasing and saturates ~ 7 s, much like the dynamics of the burst rate (black line, *right-hand ordinate*, see Fig. 1*E*).

the network's burst rate. We conclude that the threshold state of individual neurons in a network carries substantial predictive information about future synchronous events in that network.

This last conclusion offers an intriguing prediction, relating neuronal dynamics to network excitability. As demonstrated in many past studies (e.g., Marom and Shahaf 2002; Wagenaar et al. 2004), applying low-frequency stimulation from a single electrode may evoke a network synchronous event; the network's responsiveness to such rare stimuli is not compromised by the presence of frequent stimulation from a different electrode (Eytan et al. 2003) and therefore may be performed on the background of the response-clamp procedure. Our results so far predict that the probability that such low-rate stimulation will evoke a burst should be correlated with the threshold state of individual neurons in the network at the onset of stimulation. To test this prediction, we performed the following experiment: first, a directly responding neuron was selected and clamped at 10 Hz as before. Occasionally, a single stimulus was delivered from a second stimulation electrode (i.e., one that is not used for the response-clamp procedure) precisely when T seconds of quiescence elapsed since a spontaneous burst was detected online (Fig. 6, *left*, $T = 2.5$ s, *right*, $T = 5$ s). This conditioned stimulation was temporally sparse (≥ 17.5 s between consecutive stimuli), hence minimizing entrainment of the network (see Fig. 7, *H–J*).

The network activity following the conditioned stimulation is bimodally distributed (see Fig. 6, *top, insets*), reflecting the sharp threshold phenomenon these networks exhibit (Eytan and Marom 2006). We therefore classified the responses as either evoked or failed bursts (using a cutoff value that is based on the above-mentioned distribution) and calculated the stimulation-triggered averages of the total network activity (Fig. 6, *top*), as well as the changes in the clamped neuron's threshold (Fig. 6, *bottom*) for each of these subsets. As can be expected based on the results shown in Figs. 1, 4, and 5, failed population bursts are preceded by high single neuron thresholds, whereas evoked bursts are preceded by relatively low neuronal threshold. Thus the threshold of individual neurons is related not only to the network's spontaneous activity patterns but also to the net-

work's global excitability status, i.e., its responsiveness to external stimulation. This experiment was repeated on three neurons in three different networks, using four different values of T (2.5, 3.33, 5, and 10 s), all producing similar observations.

Controls and Validation

The novelty of the experimental paradigm and the analyses presented in this work require that extra care be taken to verify that the results do not reflect an artifact of our methodology. In this section we shall focus our attention on two issues: 1) the effects of repetitive high-rate stimulation on network activity; and 2) the (in)dependence of the long-term threshold effects reported here on the parameters of the controller.

Due to its nonlocal nature, the extracellular stimulation used here might affect other neurons in the network. It was therefore necessary to check these nonlocal effects on features of networkwide synchrony. It should first be noted that a time-scale separation of more than an order of magnitude exists between the stimulation rate (10 Hz for each neuron clamped) and the rate of network synchronous events (once every several seconds). Thus the response-clamp stimulation cannot be said to entrain the network's activity. While some statistical features of network synchrony are often altered by this high-rate stimulation, these alterations are variable (Fig. 7); in any event, synchrony remains the predominant feature of the network's behavior.

We now focus on validating the key feature analyzed in this study, namely the slow relaxation of the neuronal threshold following network events, a relaxation that extends until the onset of subsequent events. One may raise an alternative explanation to this result that goes as follows: 1) the decrease in the threshold during a burst is caused by the response clamp falsely identifying network-driven spikes as evoked ones, and 2) the increase in stimulation amplitude following the burst reflects compensatory effects of the response-clamp controller to this initial decrease, rather than actual increase in neuronal threshold. To overcome the former concern, the window of spike detection was narrowed down to only 4 ms around the expected spike latency (see MATERIALS AND METHODS), thus

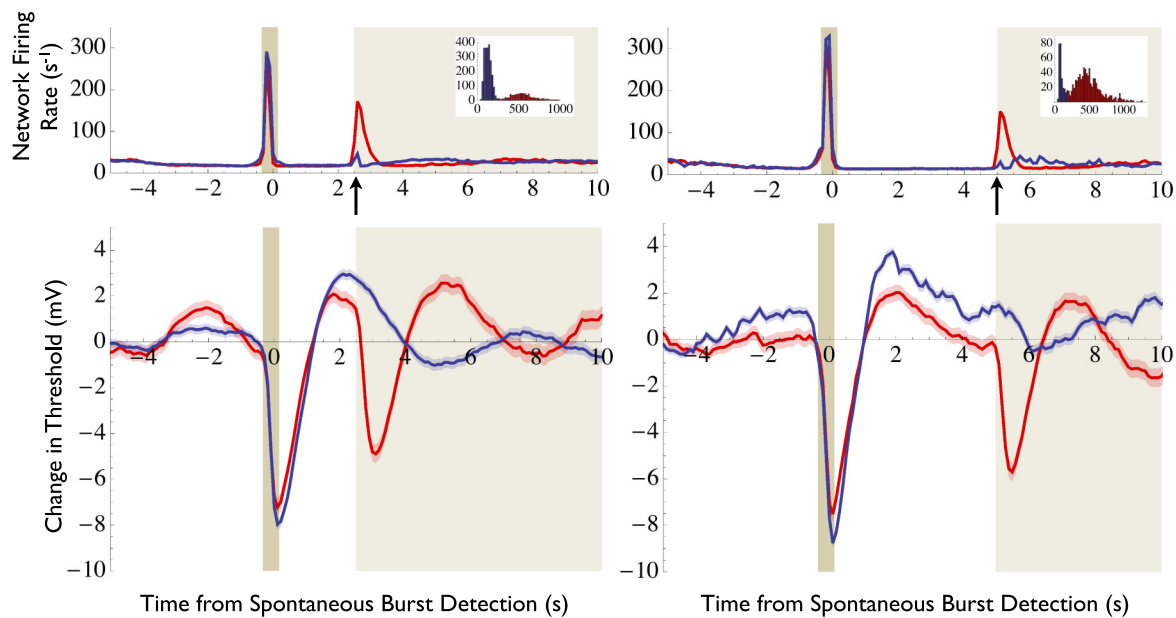


Fig. 6. Threshold dynamics are correlated with network excitability. Single neuron threshold was recorded in an active network. Occasionally, a voltage stimulus was applied to the network from a different electrode than the one used for the clamp procedure. Stimuli (depicted by arrows) were given 2.5 s (left, 3,045 stimuli in total) or 5 s (right, 2,562 stimuli in total) after the detection of a spontaneous burst. Distribution of the total network activity within 0.5 s following the stimulation time is bimodal (insets), reflecting the “all-or-none” like threshold phenomenon in the network responsiveness. Responses were therefore categorized as either evoked (red) or failed (blue) bursts. Stimulus-triggered averages of network activity (top) and threshold (bottom) dynamics in the 2 types of responses are shown. As expected given the results of Figs. 1, 4, and 5, the responsiveness of the network to a given stimulus is correlated with the threshold state of the single neuron just before the stimulus is applied. In other words, failed responses are preceded by higher threshold levels. Shaded area depicts SE. Note the threshold deflection following stimulation (present only when a burst is evoked).

minimizing the probability of false positives. Moreover, the analysis presented in Fig. 4B clearly demonstrates that the long-term effects on the threshold cannot be attributed to the intensity of neuronal firing during the burst.

As for the second concern mentioned above, if indeed the long-term dynamics measured after network events reflect the behavior of the controller (rather than intrinsic neuronal processes), these dynamics must be sensitive to how the controller is configured, namely to the parameter setting of the control algorithm. To assess how sensitive the recording is to the parameter values of the response-clamp controller, we performed the following control experiments: a neuron was clamped using two differently configured controllers: one that uses a fast time constant of response probability estimation ($\tau = 10$ s) and one that used a slow one ($\tau = 20$ s). The two controllers stimulated the neuron at 5 Hz each alternately using the same stimulation electrode (i.e., total stimulation rate was 10 Hz). Each controller received feedback only from the responses to its own stimulation. Thus one controller clamped the even responses, while the other clamped the odd ones. The rationale for this “double clamp” setup was to acquire two simultaneous recordings of the same process using two differently configured devices, each seeing different samples of that process. The resulting burst-triggered averages of change in threshold are shown in Fig. 8A. As expected, the slow controller yields a low-pass filtered version of the signal relative to the fast controller (see MATERIALS AND METHODS), expressed mainly as an attenuation of the first, fast phase of the deflection. In contrast, both the magnitude and timescale of the slower phase of the deflection are practically identical. The double clamp procedure was repeated with controllers differing in the value of either g_P (Fig. 8B) or g_I (Fig. 8C) to observe the influence of

these parameters on the averaged signal. As expected, higher gain parameters yielded a stronger averaged signal. Normalization to the peak deflection value (see Fig. 8, insets) reveals that in all other respects the recorded signals are identical.

A similar approach was exercised to expose the sensitivity of our measurement to the actual source (i.e., physical location) of the stimulation. To that end, two identical controllers were used, each stimulated the neuron using a different stimulation electrode. Here, too, the threshold deflections differed only in amplitude (Fig. 8D, inset, normalized signals). We conclude that the duration of deflection in neuronal threshold due to network synchronous activity is insensitive to both the configuration of the control algorithm and the choice of stimulation source, and therefore reflects intrinsic neuronal processes. Regarding the amplitude of the measured signal, which is evidently influenced by the experimental procedure: The dependency of this amplitude on the gain parameters imply that the “true” amplitude of the phenomenon (i.e., the amplitude at “infinite” gain) is much larger than is actually measured. However, increasing the gain parameters beyond certain values leads to loss of stability and oscillations. Thus, unlike the voltage-clamp technique, which operates at high feedback gain regimes, the response-clamp measurements can only provide attenuated and filtered versions of the “real” signal. Whatever the actual amplitude of this real signal is, the results presented in Figs. 4 and 5 clearly indicate that it carries substantial functional information about the state of both the neuron and the network.

DISCUSSION

In this work, we utilized the response-clamp methodology to investigate interactions between two levels of biological orga-

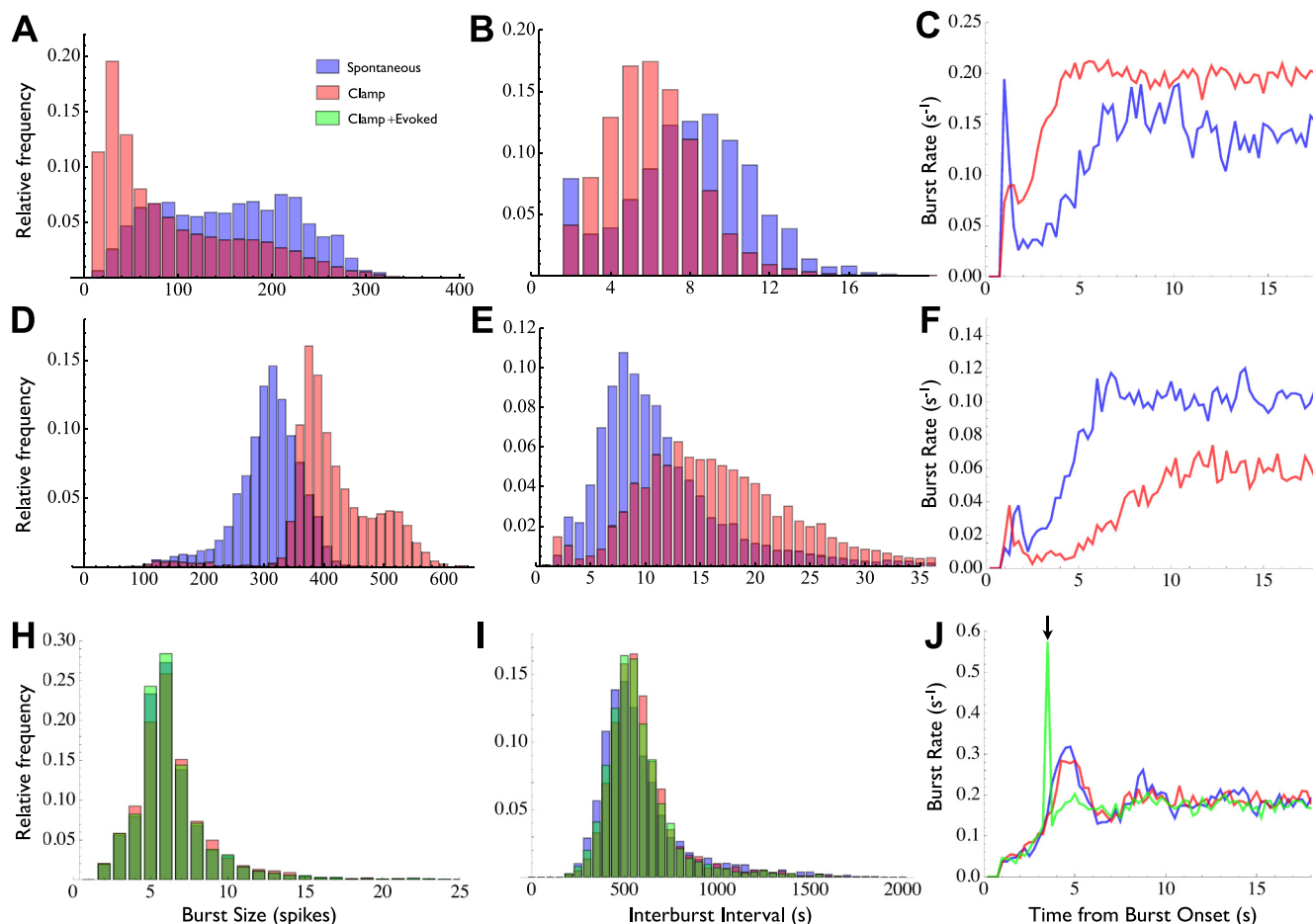


Fig. 7. Effects of response-clamp stimulation and evoked bursts on network activity statistics. Burst size distribution (A, D, and H; total number of spikes from 0.1 s before to 0.5 s after burst onset), interburst interval distribution (B, E, and I) and burst-triggered burst rate dynamics (C, F, and J) during spontaneous recordings (blue), response-clamp stimulation (red), and clamped-evoked stimulation (green; from the experiment presented in Fig. 6). Each row presents results from a different culture. In the A–C, bursts under stimulation were smaller (A), more frequent (B), and the burst rate recovery was faster (C) relative to the nonclamp control. In contrast, in the preparation shown in D–F bursts were larger (D), more sparse (E), and showed slower recovery (F) relative to the control. In H–J, the statistics of a network used for the evoked bursts experiment presented in Fig. 6 are shown. In this instance, both burst magnitude and interburst interval distributions are not markedly affected by the frequent stimulation of the response clamp nor altered when bursts were evoked by rare stimulation from a second electrode (green). Note, however, that the presence of the evoked bursts produces a sharp peak in the burst-triggered burst rate (depicted by an arrow at 3.33 s, the interval between spontaneous bursts and network stimulation in this case).

nization: the network and the single neuron. Specifically, we documented changes in cellular threshold that accompany the occurrence of a network synchronous event. We also observed similar, albeit much faster, dynamics in the properties of the spikes generated by the neuron (Fig. 3). We found that the effects of a single network synchrony event on any single neuron threshold last several seconds and are related to the time-scale of network level dynamics (Figs. 1E, 2, and 5). We further demonstrated that the long lasting shift in the threshold is input rather than output dependent (Fig. 4): the increase in the single neuron threshold following a network burst is correlated with network activity levels during the burst, rather than by the actual spikes that the neuron generates during that burst. Moreover, the magnitude of network activity during a synchronous event is correlated with the threshold state of individual neurons at the event's onset. Finally, the threshold recovery was shown to be correlated both with the network's likelihood of spontaneously generating subsequent bursts (Fig. 5), as well as with its likelihood of responding to external stimulation (Fig. 6).

These results, enabled by our capacity to monitor threshold dynamics noninvasively over extended timescales, bring up a whole range of questions rather than providing definitive answers: Which mechanisms might be envisioned as a basis of such long lasting (seconds) threshold dynamics? What makes these threshold dynamics correlated amongst the neurons, independent of their individual firing rates? How can individual neurons' threshold deflections before network synchrony entail synchrony (burst) magnitude? While an attempt to provide answers to these questions based on the experimental results presented here would be presumptuous, we do have some thoughts as to where answers might reside.

We start by acknowledging the fact that what we refer to as “dynamic threshold” might reflect a wide range of previously reported factors, which were shown over the years to have the capacity to relax at the seconds timescale. Many studies, for instance, have focused on the so-called subthreshold fluctuations of intracellular membrane potential (Lampl and Yarom 1993), which reflect, among other factors, changes in the activation of ion channels (including ligand-gated synaptic

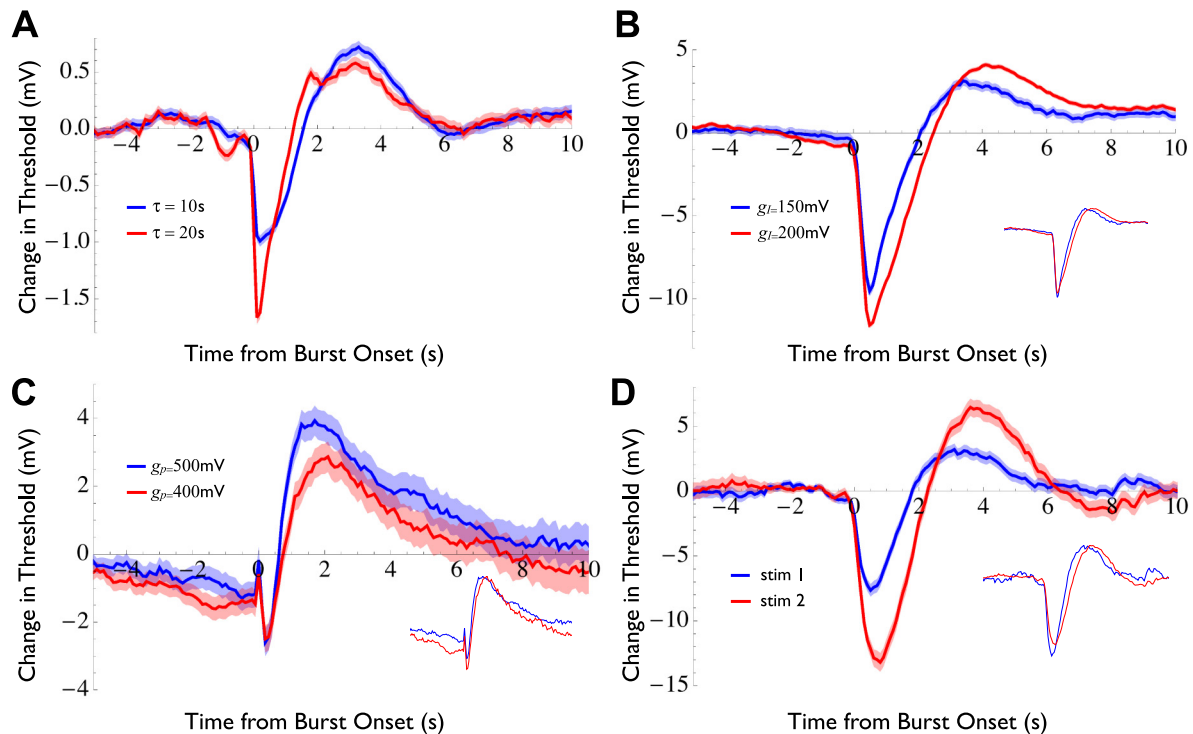


Fig. 8. Control experiments designed to verify the authenticity of the main finding. *A*: “double-clamp” procedure was performed using one stimulation electrode and 2 controllers, one with a fast estimation time constant (10 s, red) and one with a slow estimation time-constant (20 s, blue). Averaged signal of the slow estimator yields smoothed version of the one obtained from the fast estimator. *B*: same procedure with controllers differing in integrator gain coefficient (blue $g_I = 150$ mV, red $g_I = 200$ mV). Higher gain yielded a slightly amplified signal; *inset*: signals normalized to the maximal deflection. *C*: same procedure, controllers differing in proportional gain (blue $g_P = 500$, red $g_P = 400$). *Inset*: normalized signals. *D*: double-clamp procedure using 2 identical controllers and 2 stimulation electrodes stimulating the same neuron. Magnitude of the averaged signal due to the different stimulation sources is different, but the dynamics are in all other respects identical (*inset*: normalized signals). Shaded area in *A–D* represent SE.

receptors) and changes in local ion concentrations. Other works have looked at the intracellular threshold, namely the membrane potential above which an action potential develops. This threshold is what theorists call a separatrix, an unstable boundary between two modes of operation that cannot be directly measured in open loop experiments. Various analyses were used to show that this threshold is dynamic (Elul and Adey 1966) and history dependent (Azouz and Gray 2000). Such changes are related (for example) to the availability of voltage-dependent ion channels (i.e., their occupancy of non-inactivated states).

In contrast, the neuronal threshold measured by the response clamp reflects the stimulus strength needed to elicit an action potential at every given moment and therefore indicates the distance between the neuron’s current state (i.e., the subthreshold potential) and the unstable separatrix (the intracellular threshold). Therefore, all the above-mentioned mechanisms (ionic channel availability or modulation, intracellular ionic concentrations, etc.), acting independently or jointly, may give rise to this functional threshold, which carries information on the instantaneous input-output characteristics of the neuron. While acknowledging the potential benefit of linking our results with such molecular processes, the approach offered here focuses on the functional threshold as a macroscopic state variable, relating it to a higher level (network) organization.

That the threshold dynamics are correlated amongst neurons reflect the fact that all the neurons in the network receive similar summed input intensity during a synchronous event and therefore fluctuations in their threshold following each burst

are expected to be highly correlated. This interpretation is further supported by our demonstration (Fig. 4) that it is the input to a neuron, rather than its own spiking response, that determines the extent of threshold deflection. Given these correlations, it is perhaps not surprising that the time to next network synchrony, as well as the network’s responsiveness to external stimulation, are entailed by the correlated relaxation of neuronal threshold from the previous synchronous event (Figs. 5 and 6). Admittedly, we have no clue how to account for the relation between threshold deflection before network synchrony and the synchrony (burst) magnitude.

More generally, we believe, a theoretical framework for network synchrony is wanted. Such a framework must take into account the capacity of the threshold dynamics of a single neuron to be (at least partially) autonomous from its own spiking activity; the latter probably requires something beyond present day reliance on simple models of spike rate adaptation (Liu and Wang 2001), which are, as our results imply, not sufficiently rich. One possible direction in this respect would be to define the threshold as a state variable of the single neuron, effectively coupled to thresholds of other neurons. The spiking activity in this picture might be defined as a population (field) state variable that impacts on and is affected by the thresholds of individual neurons. The expansive theoretical results on emergent synchronization in weakly coupled elements, e.g., the Kuromoto model (Golomb et al. 1992; Pikovsky et al. 1996; Strogatz 2000), might serve as a good starting point for the construction of such a framework.

Finally, the mapping presented here between the two levels of organization, population and single neuron, suggests that no one level should be considered as “driving” the system; instead, and not unlike other examples in physiology (Noble 2008), the relations between the two levels might be thought of in terms emphasizing their interactive, bidirectional nature.

ACKNOWLEDGMENTS

We thank Erez Braun and Daniel Dagan for useful comments and suggestions.

GRANTS

The research leading to these results has received funding from the European Unions Seventh Framework Program FP7 under Grant Agreement 269459 and was also supported by a grant of the Ministry of Science and Technology of the State of Israel and MATERA Grant Agreement 3-7878.

DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the author(s).

REFERENCES

- Abeles M. *Corticonics: Neural Circuits of the Cerebral Cortex*. Cambridge, UK: Cambridge Univ Press, 1991.
- Azouz R, Gray C. Dynamic spike threshold reveals a mechanism for synaptic coincidence detection in cortical neurons in vivo. *Proc Natl Acad Sci USA* 97: 8110–8115, 2000.
- Beggs J, Plenz D. Neuronal avalanches in neocortical circuits. *J Neurosci* 23: 11167–11177, 2003.
- Bonifazi P, Goldin M, Picardo M, Jorquera I, Cattani A, Bianconi G, Represa A, Ben-Ari Y, Cossart R. Gabaergic hub neurons orchestrate synchrony in developing hippocampal networks. *Science* 326: 14–19, 2009.
- Droge M, Gross G, Hightower M, Czisny L. Multielectrode analysis of coordinated, multisite, rhythmic bursting in cultured CNS monolayer networks. *J Neurosci* 6: 1583–1592, 1986.
- Eckmann JP, Moses E, Stetter O, Tlustý T, Zbinden C. Leaders of neuronal cultures in a quorum percolation model. *Front Comput Neurosci* 4: 132, 2010.
- Elul R, Adey W. Instability of firing threshold and “remote” activation in cortical neurones. *Nature* 212: 1424–1425, 1966.
- Eytan D, Brenner N, Marom S. Selective adaptation in networks of cortical neurons. *J Neurosci* 23: 9349–9356, 2003.
- Eytan D, Marom S. Dynamics and effective topology underlying synchronization in networks of cortical neurons. *J Neurosci* 26: 8465–8476, 2006.
- Gal A, Eytan D, Wallach A, Sandler M, Schiller J, Marom S. Dynamics of excitability over extended timescales in cultured cortical neurons. *J Neurosci* 30: 16332–16342, 2010.
- Giugliano M, Darbon P, Arsiero M, Lüscher HR, Streit J. Single-neuron discharge properties and network activity in dissociated cultures of neocortex. *J Neurophysiol* 92: 977–996, 2004.
- Golomb D, Hansel D, Shraiman B, Sompolinsky H. Clustering in globally coupled phase oscillators. *Phys Rev A* 45: 3516–3530, 1992.
- Ham M, Bettencourt L, McDaniel F, Gross G. Spontaneous coordinated activity in cultured networks: analysis of multiple ignition sites, primary circuits, and burst phase delay distributions. *J Comput Neurosci* 24: 346–357, 2008.
- Hodgkin A, Huxley A, Katz B. Measurement of current-voltage relations in the membrane of the giant axon of Loligo. *J Physiol* 116: 424–448, 1952.
- Jimbo Y, Robinson H, Kawana A. Simultaneous measurement of intracellular calcium and electrical activity from patterned neural networks in culture. *IEEE Trans Biomed Eng* 40: 804–810, 1993.
- Kermany E, Gal A, Lyakhov V, Meir R, Marom S, Eytan D. Tradeoffs and constraints on neural representation in networks of cortical neurons. *J Neurosci* 30: 9588–9596, 2010.
- Lampl I, Yarom Y. Subthreshold oscillations of the membrane potential: a functional synchronizing and timing device. *J Neurophysiol* 70: 2181–2186, 1993.
- Levine W (editor). *The Control Handbook*. Boca Raton, FL: CRC, 1996.
- Lewicki MS. A review of methods for spike sorting: the detection and classification of neural action potentials. *Network* 9: R53–R78, 1998.
- Li C, Poo M, Dan Y. Burst spiking of a single cortical neuron modifies global brain state. *Science* 324: 643–646, 2009.
- Liu Y, Wang X. Spike-frequency adaptation of a generalized leaky integrate-and-fire model neuron. *J Comput Neurosci* 10: 25–45, 2001.
- Maeda E, Robinson H, Kawana A. The mechanisms of generation and propagation of synchronized bursting in developing networks of cortical neurons. *J Neurosci* 15: 6834–6845, 1995.
- Marom S, Shahaf G. Development, learning and memory in large random networks of cortical neurons: lessons beyond anatomy. *Q Rev Biophys* 35: 63–87, 2002.
- Noble D. *The Music of Life: Biology Beyond Genes*. Oxford, UK: Oxford Univ. Press, 2008.
- Pikovsky A, Rosenblum M, Kurths J. Synchronization in a population of globally coupled chaotic oscillators. *Europhys Lett* 34: 165, 1996.
- Robinson H, Kawahara M, Jimbo Y, Torimitsu K, Kuroda Y, Kawana A. Periodic synchronized bursting and intracellular calcium transients elicited by low magnesium in cultured cortical neurons. *J Neurophysiol* 70: 1606–1616, 1993.
- Singer W. Neuronal synchrony: a versatile code review for the definition of relations? *Neuron* 24: 49–65, 1999.
- Stegenga J, Le Feber J, Marani E, Rutten W. Analysis of cultured neuronal networks using intraburst firing characteristics. *IEEE Trans Biomed Eng* 55: 1382–1390, 2008.
- Strogatz S. From Kuramoto to Crawford: exploring the onset of synchronization in populations of coupled oscillators. *Physica D* 143: 1–20, 2000.
- Van Pelt J, Corner M, Wolters P, Rutten W, Ramakers G. Longterm stability and developmental changes in spontaneous network burst firing patterns in dissociated rat cerebral cortex cell cultures on multielectrode arrays. *Neurosci Lett* 361: 86–89, 2004.
- Wagenaar D, Pine J, and Potter S. Effective parameters for stimulation of dissociated cultures using multi-electrode arrays. *J Neurosci Meth* 138: 27–37, 2004.
- Wagenaar D, Pine J, Potter S. An extremely rich repertoire of bursting patterns during the development of cortical cultures. *BMC Neurosci* 7: 11, 2006.
- Wallach A, Eytan D, Gal A, Zrenner C, Marom S. Neuronal response clamp. *Front Neuroeng* 4: 3, 2011.
- Zrenner C, Eytan D, Wallach A, Thier H, Marom S. A generic framework for real-time multi-channel neuronal signal analysis, telemetry control and sub-millisecond latency feedback generation. *Front Neurosci* 4: 173, 2010.